CMSC 420: Preliminaries

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February 8, 2026

1 Problems **2**

1 Problems

1. Calculate and simplify.

Sum	**Result**
(a) $\sum_{i=5}^{105} 7$	
(b) $\sum_{i=5}^{105} i$	
(c) $\sum_{i=n}^{2n} i$ with $n \geq 0$	
(d) $\sum_{i=1}^{117} i^2$	
(e) $\sum_{i=0}^{7} \left[1 + 2^{3i}\right]$	

2. Suppose that we have a 0-indexed list `A` of length `n`. We have the following pseudocode:

```
function shift(A,n,i):
    for j = i to n-2 inclusive:
        A[j] = A[j+1]
    end for
end function
```

Suppose the code `A[j] = A[j+1]` takes j seconds to execute. If an index $0 \leq i \leq n-1$ is chosen uniformly at random what is the expected time it will take for the code to run?

Note: There are several $\sum$ expressions involved. You should show what these evaluate to but after that you can use software tools to simplify the resulting expressions.

3. Suppose A is a 0-indexed $n \times n$ array. We choose a coordinate (i, j) with all coordinates equally likely. We delete all elements in row i and all elements in column j. When we do this, all elements in the lower left (if any) shift up (one shift per element), all elements in the upper right (if any) shift left (one shift per element), and all elements in the lower right (if any) shift up and left (two shifts per element). If each shift takes one second how long do we expect the deletion to take? Simplify your final (polynomial) expression.

4. Solve the following recurrence relation by digging down:

$$T(n) = 2T(n-1) + n \quad \text{with } T(0) = 3$$

5. Simplify the following sum:

$$1 + \sum_{i=0}^{h-1} 2 \lceil m/2 \rceil^i \left(\lceil m/2 \rceil - 1 \right)$$

6. It is the case that $\mathcal{O}(\sqrt{n})$ is slower than $\mathcal{O}(\lg n)$. Formally this means that we have $\lg n = \mathcal{O}(\sqrt{n})$ but $\sqrt{n} \neq \mathcal{O}(\lg n)$.

   (a) State the limit theorem (from CMSC351) which applies to $\mathcal{O}$.

   (b) Prove that $\lg n = \mathcal{O}(\sqrt{n})$ using the $\mathcal{O}$ limit theorem.

(c) Prove that $\sqrt{n} \neq \mathcal{O}(\lg n)$ using the definition of $\mathcal{O}$.
Note: This is a short but fairly challenging mathematical argument.

7. Suppose h is a nonnegative integer and we are given that:

$$n \leq \sum_{i=0}^{h} 3^i$$

Find an integer lower bound for h which does not include a sum.

8. When we talk about *the Ackermann function* in the context that we do, we generally mean the function defined first using:

$$A(0, n) = n + 1$$
$$A(m + 1, 0) = A(m, 1)$$
$$A(m + 1, n + 1) = A(m, A(m + 1, n))$$

Then via:

$$Ack(n) = A(n, n)$$

Calculate $Ack(n)$ for $n = 0, 1, 2$. Show intermediate steps, don't just give the answer.

9. Suppose algorithm A has time complexity $T_A(n) = \Theta(n)$ and algorithm B has time complexity $T_B(n) = \Theta(n^2)$.

   (a) Which algorithm is generally accepted as "faster" and why?

   (b) Give an example of two specific running times $T_A(n)$ and $T_B(n)$ such that algorithm B is faster for $0 \leq n < 10000$ and algorithm A is faster for $n > 10000$ and show why this is the case. Along with your example explain how you came up with it, so don't simply give the example. Both should be valid for all $n \geq 0$.

10. We wish to store a collection of n key-value pairs where the keys are integers and the values are strings. We put the key-value pairs in a 0-indexed list of objects where each object represents a pair and such that the keys are in increasing order. For example the following key-value pairs:

$$[3, apple],\ [10, cat],\ [5, math],\ [-2, hello]$$

Would be stored in a list A with:

$$A[0] = [-2, hello],\ A[1] = [3, apple],\ A[2] = [5, math],\ A[3] = [10, cat].$$

(a) To search for a particular key we will use a binary search. What is the best and worst case $\mathcal{O}$ time complexity of this? Explain.

(b) To insert a key-value pair we use a binary search to find the next smallest key (do you know how to do this?) and shift to make space accordingly. What is the best and worst case $\mathcal{O}$ time complexity of this? Explain.

(c) To delete a key-value pair we use a binary search to find the key and then splice to remove the key-value pair accordingly. What is the best and worst case $\mathcal{O}$ time complexity of this? Explain.

11. One of the most frustrating things when it comes to proofs related to algorithms is the fact that floors and ceilings show up and get in the way. Prove that for all $x \geq 2$ we have:

$$\left\lfloor \log_{3/2}(x+1) \right\rfloor - \left\lfloor \log_{3/2} x \right\rfloor \leq 1$$

Note 1: This is not particularly difficult if you're comfortable with some basic calculus so if you find your solution getting a little out of hand you're probably doing more than you need to!

12. If algorithm A has time complexity $\mathcal{O}(n \lg n)$ and algorithm B has time complexity $\mathcal{O}(n^2)$ does this mean we should always use algorithm A? Explain in no more than one paragraph!

13. Suppose A is a 0-indexed list of integers which has the property that if A has length j then inserting an integer at index $0 \leq k \leq j$ (where $k = j$ is an append) takes time $j - k + 1$ seconds.

    (a) Starting with a list of length 0, how many seconds will it take to append n integers?

    (b) Starting with a list of length n, if an integer is inserted into a random index $0 \leq k \leq n$ (with each index equally likely) what is the expected amount of time required? Simplify your answer.

14. For each of the following recurrence relations with base cases given, if we dig down k steps calculate which k (as an expression involving n) would yield the base case. The assumption on n is given which guarantees the base case is always reached.

Relation	Assumption on n	k expression
$T(n) = 2T(n/3) + n$ with $T(1) = 3$	$n \in \{1, 3, 9, 27, ...\}$	
$T(n) = 3T(n-3)$ with $T(0) = 5$	$n \in \{0, 3, 6, 9, 12, ...\}$	
$T(n) = T(n-1) + n^2 + n$ with $T(2) = 7$	$n \in \{2, 3, 4, ...\}$	
$T(n) = 2T(n-2)$ with $T(1) = 10$	$n \in \{1, 3, 5, 7, 9, ...\}$	
$T(n) = aT(n/b) + cn$ with $T(1) = 2$	$n \in \{1, b, b^2, b^3, ...\}$	
$T(n) = 4T(n/2 - 1)$ with $T(0) = 17$	$n \in \{2, 6, 14, 30, ...\}$	

15. Calculate each of the following permutations or combinations and simplify.

Problem	Answer
$P(10, 3)$	
$C(10, 3)$	
$P(7, 7)$	
$C(7, 7)$	

16. For each of the following, some fact about a recurrence relation is given. From that fact what $\mathcal{O}$, Ω, or Θ fact can you conclude about $T(n)$? Draw the strongest conclusion possible.

Fact	Conclusion about $T(n)$
(a) $T(n) = n \lfloor \lg n + 10 \rfloor$	
(b) $T(n) = 4T(n/4) + n$	
(c) $T(n) \geq n^2 + 2n + 1$	
(d) $T(n) \leq \frac{1}{n}$	
(e) $T(n) \leq n\lvert \sin(n) \rvert$	

17. Calculate each of the following and simplify:

Calculation	Result
(a) $\sum_{i=1}^{10} i$	
(b) $P(8,2)$	
(c) $C(8,2)$	

18. Suppose that we have a 0-indexed list `A` of length `n`. We have the following pseudocode:

```
function zero(A,n,i):
    for j = 0 to i inclusive:
        A[j] = 0
    end for
end function
```

Suppose the code `A[j] = 0` takes j seconds to execute. If an index $0 \leq i \leq n-1$ is chosen uniformly at random what is the expected time it will take for the code to run? You can (should) leave any sums un-evaluated in your answer.

Time =	

19. For each of the following, some fact about a function $T(n)$ is given. From that fact what $\mathcal{O}$, Ω, or Θ fact can you conclude about $T(n)$? Draw the strongest conclusion possible.

Answers should have the form $\mathcal{O}(g(n))$ or $\Theta(g(n))$ or $\Omega(g(n))$ for some nice function $g(n)$.

Fact	Conclusion about $T(n)$
(a) $T(n) \leq 2n \lg n + n + 1$	
(b) $T(n) = 2T(n/2) + n + 1$	
(c) $T(n) \geq 0.01n + \sqrt{n}$	
(d) $T(n) = 3T(n/27) + \sqrt{n}$	
(e) $T(n) = 3n + \lfloor n/10 \rfloor (\lceil n/10 \rceil + 1)$	

20. Suppose n is a positive integer. Find an expression in terms of n for the...

Criteria	Result
...largest $k \in \mathcal{Z}$ Such That $2^k \leq n$	
...smallest $k \in \mathcal{Z}$ Such That $2^{k+1} \geq n$	
...largest $k \in \mathcal{Z}$ Such That $\log_2 k \leq n$	

21. Suppose you roll a 6-sided die until a 6 appears and then you count the number of rolls which occurred before this 6 appeared.

    (a) What is the probability that there were no rolls before the first 6?

    (b) What is the probability that there was exactly one roll before the first 6?

    (c) What is the probability that there were either two or fewer rolls before the first 6?

    (d) What is the probability that there were three or more rolls before the first 6?

22. Solve the following recurrence relation by digging down:

$$T(n) = T(n/2) + 1 \quad \text{with } T(1) = 3$$

23. Consider the following recurrence relation:

$$T(n) \leq T(n-2) + 1 \quad \text{with } T(0) \leq 10 \text{ and } T(1) \leq 20$$

(a) Dig down to draw a conclusion about $T(n)$ for the case where n is even.

(b) Dig down to draw a conclusion about $T(n)$ for the case where n is odd.

(c) What $\mathcal{O}$ fact can we conclude about $T(n)$?

24. Consider the following recurrence relation:

$$T(n) \leq 2T(n-3) + 1 \quad \text{with } T(0) = 1 \text{ and } T(1) = 2 \text{ and } T(2) = 3$$

Dig down to draw a conclusion about $T(n)$ for the case where n is a multiple of 3.

25. Suppose n is an unknown positive integer and you wish to add all powers of 2 which are less than or equal to n

   (a) Write down a closed form for this sum.

   (b) What nice $\mathcal{O}$ fact can you conclude about this?

26. Calculate the following sum, showing steps:

$$\sum_{i=1}^{10} 5^{6-i}$$

27. Calculate the following sum, showing steps:

$$\sum_{i=3}^{n} 7^{2i+3}$$